

CrossPath: Unlocking Off-Diagonal Compatibility for Composed Image Retrieval

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Abstract

Composed image retrieval combines a reference image and a text modification to search a gallery. Existing ensembles average only each system’s matched query-gallery score, leaving cross-system pairings unused. We introduce CrossPath, which evaluates the complete 2×2 compatibility matrix between two frozen retrieval endpoints. Its off-diagonal paths exchange query and gallery encoders, and their mean adds no fusion parameters. For datasets with cutoff-specific path preferences, a lightweight listwise router selects a rank path with exact fallback to the base endpoint. On FashionIQ, heterogeneous DQU and MCoT endpoints reach 64.58 R@10 and 82.82 R@50, gains of 1.02 and 0.48 points over same-pipeline MCoT. On FashionGen, joint routing improves a strong reproduced endpoint by 1.36 R@1, 1.10 R@5, and 0.53 R@10, and exceeds a compute-matched diagonal ensemble. On CIRR, conservative cross-path fusion with query-conditioned target refinement raises Avg from 81.64 to 82.38, extending the gain to open-domain retrieval. A coordinate-scrambling control preserves within-endpoint rankings yet collapses off-diagonal retrieval, showing that trained CIR systems contain reusable ability in the paths between endpoints.

1. Introduction

Composed image retrieval (CIR) turns an image and a natural-language modification into a precise visual search query. A user can provide a garment and ask for a version with a different color, texture, or silhouette. The system must preserve the relevant visual content, apply the requested change, and retrieve the intended target from a large gallery. This combination of visual retention and linguistic editing makes CIR a central test of multimodal representation learning [21, 32, 36].

Modern CIR methods have steadily improved the query representation. Contrastive composition models align learned multimodal query with target images [1]. Raw data fusion lets a pretrained vision-language model process image-derived text and text-rendered images [36].

More recent systems use large multimodal models, synthetic data, explicit conflict handling, or multi-level reasoning [9, 12, 30, 33]. Their mechanisms differ, but their retrieval interface is uniform: endpoint i emits a query vector q_i and gallery vectors g_i , then ranks candidates with $q_i^\top g_i$.

This interface creates an overlooked opportunity. Consider two endpoints trained for the same CIR task. Conventional ensembling averages their matched scores, $q_0^\top g_0$ and $q_1^\top g_1$. It never evaluates $q_0^\top g_1$ or $q_1^\top g_0$. When the endpoints remain in a compatible pretrained vision-language space [23], these off-diagonal terms are meaningful retrieval functions. They also need not duplicate the behavior of either endpoint. A query composer can improve which semantics are retained, while a gallery encoder can improve how those semantics are indexed. Exchanging the two sides creates a path that no single endpoint exposes.

We introduce CrossPath to expose this complete compatibility space. As shown in Fig. 1, two endpoints define a 2×2 score matrix. The cross mean combines its off-diagonal entries with no trainable fusion parameter. It can be applied directly when both endpoints share a coordinate-compatible embedding space, including heterogeneous CIR models built on the same pretrained representation. We also develop a routed variant for datasets whose evaluation cutoffs prefer different paths. The router operates on deterministic rank-percentile trajectories, scores only candidates whose top- k membership can change, and falls back exactly to the base ranking unless the predicted utility clears a calibrated threshold.

The empirical pattern is consistent across FashionIQ [36], FashionGen [40], and the open-domain CIRR benchmark [21]. On FashionIQ, combining the query side of MCoT-MVS [9] with DQU-CIR [35] gallery representations gives the strongest individual off-diagonal paths. Their cross mean improves both the strongest same-pipeline endpoint and a compute-matched diagonal ensemble. On FashionGen, the cross mean yields a large R@1 gain, while joint routing retains its early-rank strength and raises R@5 and R@10. CIRR contains a larger gap between the two endpoints, so a conservative one-sided cross path is more effective than an equal-weight mean. Coupling this path with

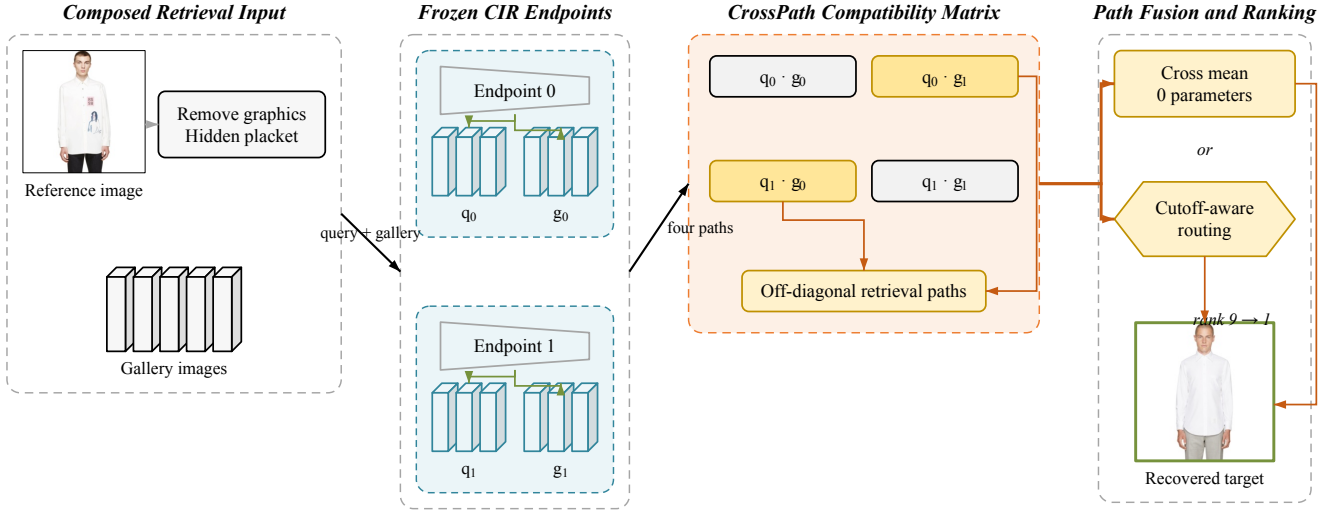


Figure 1. **CrossPath** exposes off-diagonal compatibility between frozen CIR endpoints. Two endpoints encode the composed query and gallery, producing four similarities $S_{ij} = q_i^\top g_j$. Standard ensembling uses only the diagonal. CrossPath also evaluates the two cross paths and returns one ranking. In the shown FashionGen query, cross-path fusion moves the ground-truth target from rank 9 to rank 1.

query-conditioned target refinement improves the MCOT validation Avg by 0.74 points. We further apply a shared signed permutation to one endpoint’s query and gallery coordinates. This operation leaves every within-endpoint dot product unchanged, but destroys cross-coordinate correspondences. Cross-path recall then falls almost to zero on FashionIQ and FashionGen. The control identifies a concrete mechanism rather than attributing the improvement to generic model averaging.

Our contributions are threefold:

- We formulate endpoint composition as a complete query-gallery compatibility matrix and identify its off-diagonal paths as a source of reusable CIR ability.
- We present a zero-parameter cross-path reducer and a lightweight cutoff-aware router that produce one standard ranking while retaining a compute-matched diagonal ensemble as the central control.
- We establish gains on FashionIQ, FashionGen, and CIRRG supported by module ablations, directional path analysis, an invariance-preserving scrambling control, efficiency measurements, and retrieval visualizations.

2. Related Work

Composed image retrieval. TIRG established image-text composition as a learned retrieval problem and introduced a gated residual fusion operator [32]. FashionIQ later provided natural-language feedback for fashion search [36], while CIRRG extended evaluation to open-domain images [21]. PinPoint broadens evaluation with explicit negatives, paraphrases, and multi-image queries [22]. Early supervised methods learned the composed query through

joint visual-semantic matching [4], visiolinguistic attention [5], modality-agnostic fusion [8], or fashion-specific compositional transitions [39]. Other approaches used residual composition [26], dual objectives [16], comprehensive linguistic-visual fusion [34], or attribute-aware contrastive learning [31]. Later systems introduced content-style modulation [17], semantic attention [13], explicit/implicit target matching [7], and multi-grained uncertainty regularization [6]. CLIP4CIR strengthened supervised composition with task-oriented CLIP features and contrastive learning [1]. DQU-CIR moved fusion to the input level through textual and visual query unification [35]. ConText-CIR aligns textual concepts with relevant image content [37], CCIN neutralizes compositional conflicts [30], and MAPNet models multi-scale schema interactions and noisy negatives [25]. CoLLM uses large language models to create and encode multimodal training queries [12]; ReCALL recalibrates retrieval ability after MLLM adaptation [38]; MCOT-MVS selects multi-level evidence through multimodal reasoning [9]. FashionMV introduces product-level multi-view retrieval and ProCIR, providing the FashionGen protocol used in this work [40].

Data-efficient and zero-shot CIR. Because annotated CIR triplets are costly, a large body of work removes or reduces task-specific supervision. Pic2Word maps an image into the language token space [24], and SEARLE learns the corresponding pseudo word through textual inversion [2]. LinCIR trains a projection using language-only self-supervision [11]. KEDs enriches pseudo-word representations with an external database and textual concept alignment [27], while CompoDiff edits image features with

a latent diffusion model [10]. RTD adapts the text encoder after pretraining [3], while HIT learns hierarchical pseudo words with text-conditioned filtering [19]. SLERP and text-anchored tuning combine image and text representations geometrically [14]. Visual Delta Generator creates semantic supervised triplets with a multimodal generator [15]. Recent approaches also predict missing target content [28], reason before retrieval [29], generate imagined proxies [18], augment retrieval with composed image generation [33], or learn under noisy triplets [20]. These methods improve the contents or training of one query representation. CrossPath instead composes already trained retrieval endpoints and is therefore complementary to advances in endpoint design.

Ensembling and representation compatibility. Score averaging is a strong retrieval baseline because independently trained models often make complementary errors. Its usual form is diagonal: each query encoder is paired only with its own gallery encoder. CrossPath asks a different question. If two systems inhabit aligned coordinates, can the query function learned by one system retrieve against the gallery geometry learned by another? The resulting cross paths preserve the standard nearest-neighbor interface, but expose directional interactions hidden by diagonal averaging. Our compute-matched diagonal ensemble separates this effect from the benefit of running two endpoints, and the coordinate-scrambling experiment separates genuine compatibility from coincidental score aggregation.

3. Method

3.1. Problem Setup

A CIR query consists of a reference image I_r and a modification text t . Given a gallery $\mathcal{G}$, the goal is to rank the target image I_t above all other candidates. Endpoint i contains a query composer C_i and a gallery encoder V_i :

$$q_i = \text{norm}(C_i(I_r, t)), \quad g_i(x) = \text{norm}(V_i(x)),$$

where $x \in \mathcal{G}$ and all vectors have dimension d . A conventional endpoint ranks x with $q_i^\top g_i(x)$. We keep all endpoint parameters frozen.

3.2. The Cross-Endpoint Compatibility Matrix

Two endpoints define four query-gallery pairings:

$$\mathbf{S}(x) = \begin{bmatrix} q_0^\top g_0(x) & q_0^\top g_1(x) \\ q_1^\top g_0(x) & q_1^\top g_1(x) \end{bmatrix} = \begin{bmatrix} S_{00}(x) & S_{01}(x) \\ S_{10}(x) & S_{11}(x) \end{bmatrix}.$$

The diagonal terms are the two original retrieval systems. Their mean,

$$D(x) = \frac{1}{2} (S_{00}(x) + S_{11}(x)),$$

is our compute-matched ensemble. It uses the same four encoded vectors as CrossPath and is the appropriate control for additional endpoint cost.

The off-diagonal entries exchange the query and gallery sides. We combine them as

$$X(x) = \frac{1}{2} (S_{01}(x) + S_{10}(x)). \quad (4)$$

This cross mean has no learned fusion parameters. It only requires aligned embedding dimensionality and meaningful coordinate correspondence. Both conditions hold for the endpoint pairs in our experiments. The use of both directions is important because compatibility is not symmetric: S_{01} and S_{10} couple different learned functions even though each score is a dot product.

3.3. Joint Rank-Path Routing

Fixed X is effective when one ranking improves all target cutoffs. FashionGen exhibits a more structured tradeoff: X is strongest at R@1, while diagonal paths are stronger at deeper cutoffs. We therefore construct a small action space that moves from the base ranking S_{00} toward either S_{11} or X .

Raw scores from independently trained endpoints need not share calibration. For any score function A , we convert each candidate to a deterministic rank percentile $p_A(x) \in [0, 1]$. Ties are resolved by ascending gallery identifier. A path action toward branch $B \in \{S_{11}, X\}$ is

$$p_{B,\alpha}(x) = (1 - \alpha)p_{S_{00}}(x) + \alpha p_B(x), \quad \alpha \in \{0, \frac{1}{8}, \dots, 1\}. \quad (5)$$

The two branches share the base action, giving $9 + 8 = 17$ rankings. For cutoff k , only candidates whose top- k membership changes along these actions can affect recall. We call this set $\mathcal{B}_k$.

The router is a two-layer listwise scorer over $\mathcal{B}_k$ plus a learned none class. For candidate x , its features contain the query, candidate embedding, elementwise interaction, absolute difference, endpoint rank percentiles, action membership trace, and normalized cutoff. The listwise target is the ground-truth candidate when it lies in $\mathcal{B}_k$, and none otherwise. This training target asks a direct question: which boundary candidate is responsible for a possible recall change?

At inference, the predicted responsibility distribution $\pi_k(x)$ determines the utility of action a :

$$U_k(a) = \sum_{x \in \mathcal{B}_k} \pi_k(x) [(1 - m_x^0)m_x^a - \lambda m_x^0(1 - m_x^a)], \quad (6)$$

where m_x^a indicates top- k membership, action 0 is the base, and $\lambda = 2$ penalizes regressions. Utilities are averaged across evaluation cutoffs so each query still emits one ranking. We select the highest-utility action only when it exceeds

a development-set threshold. Otherwise the output is exactly the base order. The router has 0.526M parameters for $d = 1024$ and hidden width 128.

3.4. Conservative Transfer and Target Refinement

Equal-weight fusion is unnecessary when one endpoint is clearly stronger. Let endpoint 1 be the stronger system. We retain its query representation and inject the alternative gallery geometry through

$$S_\alpha(x) = (1 - \alpha)S_{11}(x) + \alpha S_{10}(x), \quad 0 < \alpha < \frac{1}{2}.$$

This form preserves the strong query composer and treats the cross path as a bounded correction. We use $\alpha = \frac{1}{4}$ for CIRR.

CIRR also requires fine discrimination among visually similar members of each six-image subset. We therefore refine the top-50 candidates with the frozen segment features available to MCoT. For segment tokens $\{z_{x,s}\}$, projected query $\hat{q}$, and attention-pooled target $\bar{z}_x$, the score is

$$\begin{aligned} a_{x,s} &= \text{softmax}_s \left((W_q q)^\top W_k z_{x,s} / \sqrt{h} \right), \\ \bar{z}_x &= \sum_s a_{x,s} W_v z_{x,s}, \\ \delta_\theta(q, x) &= f_\theta[\hat{q} \odot \bar{z}_x; \hat{q}; \bar{z}_x], \\ \tilde{S}(x) &= S_\alpha(x) + \delta_\theta(q, x). \end{aligned}$$

The final layer of f_θ is zero initialized, so training starts from the frozen ranking. The 1.18M-parameter head learns only from CIRR training queries with endpoint top-ranked negatives and subset members. All endpoint and segmentation parameters remain frozen.

3.5. Why Off-Diagonal Paths Can Help

Endpoint training jointly changes two objects: the query composition function and the gallery geometry. The diagonal score entangles their effects. Equation 2 separates them. C_1 extracts a more faithful modification intent while V_0 retains a more discriminative gallery organization, S_{10} can surpass both S_{00} and S_{11} . The reverse direction can behave differently. CrossPath captures both possibilities without retraining either endpoint.

This view also gives a sharp intervention test. Let P be a signed permutation matrix and replace q_1, g_1 by Pq_1, Pg_1 . Since $P^\top P = I$, $q_1^\top g_1$ is unchanged, as are all rankings of endpoint 1 and the diagonal ensemble. In contrast, $q_0^\top Pg_1$ and $(Pg_1)^\top g_0$ lose coordinate correspondence. A large cross-path drop under this transformation is direct evidence that the original gain depends on compatible coordinates

4. Experiments

4.1. Experimental Setup

Datasets and metrics. FashionIQ contains three categories, dress, shirt, and toptee, and evaluates retrieval from a reference garment and two natural-language modifications [36]. Following the directly comparable validation protocol, we exclude the source image and report the category mean of R@10 and R@50. FashionGen is evaluated through the FashionMV validation protocol [40], with 9,031 composed queries and 5,292 gallery products. We report R@1, R@5, and R@10. CIRR contains natural images from diverse domains and evaluates a reference image paired with one relative caption [21]. We report global R@1, R@5, R@10, and R@50; subset recall at ranks 1, 2, and 3; and Avg = (R@5 + subset R@1)/2.

Endpoints. For FashionIQ, our primary pair combines the author-released MCoT-MVS endpoint [9] with reproduced DQU-CIR endpoints [35]. We evaluate DQU Base and a GradCache effective-batch-128 checkpoint. Every local method uses the same gallery convention and stable source exclusion. For FashionGen, we use two independently trained 0.8B ProCIR-compatible A1 endpoints based on the public FashionMV code [40]. CIRR pairs the released MCoT endpoint with a DQU endpoint trained from the authors' recipe. The diagonal ensemble and CrossPath always encode the same two queries and two gallery representations.

Implementation. Embeddings are ℓ_2 normalized before scoring. The FashionIQ cross mean is fully parameter free. The FashionGen router uses 17 actions, hidden width 128, three epochs, and regression cost $\lambda = 2$. Training and threshold selection use only the internal split; the reported validation ranking is produced by the frozen router. On CIRR, the target-refinement head reranks the top 50 global candidates and the subset members. It uses eight segment tokens per image, hidden width 512, and eight training epochs; the best frozen checkpoint is followed by the fixed $\alpha = 0.25$ score in Eq. 7. All recall values are percentages. Gallery encoding is offline, and experiments run on one RTX 4090.

4.2. Main Results

Table 1 shows that the reproduced A1 endpoint exceeds the published ProCIR result by 4.26 R@5 and 2.45 R@10. Running a second endpoint and averaging only the diagonal gives modest gains. The cross mean raises R@1 from 42.73 to 44.10, exceeding the diagonal ensemble by 1.25 points. Joint routing retains almost all of this early-rank gain and reaches 80.36 R@5 and 88.28 R@10. Relative to the base, the improvement is 1.36, 1.10, and 0.53 points. Paired bootstrap confidence intervals for these gains are [0.96, 1.68], [0.84, 1.34], and [0.30, 0.74], respectively.

Table 1. **FashionGen results under the FashionMV validation protocol.** Public baselines are reported by FashionMV [40]. The diagonal ensemble is compute matched to our method. Best local results are bold.

Method	Venue/src.	R@1	R@5	R@10
CLIP4CIR	CVPR'22	–	17.10	25.00
SPRC	FMV'26	–	42.70	53.00
Qwen3-VL-2B	FMV'26	–	63.00	74.10
Qwen3-VL-8B	FMV'26	–	74.70	83.50
Doubao-E-V	FMV'26	–	67.20	77.10
ProCIR	arXiv'26	–	75.00	85.30
A1 reproduced base	Local	42.73	79.26	87.75
Diagonal ensemble	Local	42.85	79.74	88.24
Cross mean (ours)	Local	44.10	79.74	88.16
CrossPath joint routing (ours)	Local	44.09	80.36	88.28

Table 2. **FashionIQ validation results.** Paper values use their published protocol. Local results are recomputed in one source-excluded pipeline. Diagonal is compute matched.

Method	Venue	R@10	R@50
TIRG [32]	CVPR'19	17.40	37.39
CLIP4CIR [1]	CVPR'22	38.40	61.74
CCIN [30]	CVPR'25	54.41	74.76
DQU-CIR [35]	SIGIR'24	62.00	81.58
MCoT-MVS [9]	WWW'26	63.24	82.01
MCoT-MVS, same pipeline	Local	63.56	82.33
Diagonal ensemble	Local	64.52	82.75
Cross mean, Base $\times$ MCoT (ours)	Local	64.61	82.57
Cross mean, GC $\times$ MCoT (ours)	Local	64.58	82.82

FashionIQ provides a heterogeneous test: DQU performs raw-data fusion, whereas MCoT uses multimodal reasoning and hierarchical visual selection. Table 2 shows that this difference is useful after training. DQU Base with MCoT obtains the highest R@10 at 64.61. Replacing DQU Base with the GradCache endpoint gives the highest R@50 at 82.82. The latter improves the same-pipeline MCoT endpoint by 1.02 R@10 and 0.48 R@50. It also exceeds the compute-matched diagonal ensemble at both cutoffs. Under the include-source convention of the released MCoT evaluator, the corresponding gain is 1.21 R@10 and 0.55 R@50, confirming that the improvement is not tied to source handling.

Open-domain transfer. Table 3 extends CrossPath beyond fashion retrieval. Conservative fusion raises CIRR Avg from 81.64 to 82.19, with gains in global recall through rank 10 and all subset metrics. Query-conditioned target refinement reaches 55.78 R@1, 85.77 R@5, and 82.38 Avg. The combined model improves the frozen MCoT endpoint by 0.37 R@1, 0.50 R@5, and 0.74 Avg. Weak fusion obtains the highest subset R@1, while target refinement improves global top ranks and subset ranks 2 and 3. These components

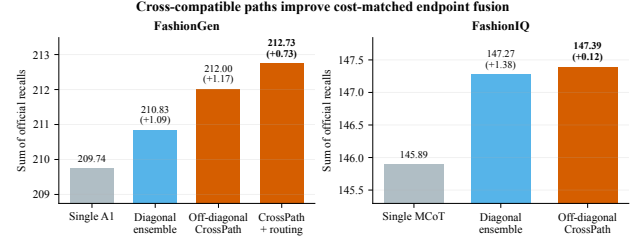


Figure 2. **Cumulative component ablation.** FashionGen uses R@1 + R@5 + R@10; FashionIQ uses R@10 + R@50. Every added component yields a positive gain.

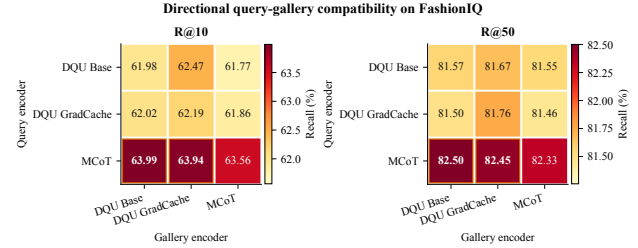


Figure 3. **Directional FashionIQ compatibility.** Each cell pairs the row query encoder with the column gallery encoder and reports R@10/R@50. Off-diagonal strength is concentrated in MCoT queries paired with DQU galleries.

mentary changes support the staged design in Sec. 3.4.

4.3. Component Analysis

Figure 2 starts from the strongest single endpoint and adds one component at a time. On FashionGen, the diagonal ensemble contributes 1.09 SumR, off-diagonal compatibility adds 1.17, and routing adds another 0.73. The final SumR improves from 209.74 to 212.73. On FashionIQ, the diagonal ensemble contributes 1.38 and replacing diagonal-only fusion with cross paths adds 0.12, reaching 147.39. The smaller final step is consistent with an already strong heterogeneous ensemble, but it is obtained without fusion training.

Directionality. Figure 3 expands the analysis to three endpoints. MCoT queries paired with DQU Base and DQU GradCache galleries reach 63.99/82.50 and 63.94/82.45. Both improve over the MCoT diagonal at 63.56/82.33. The reverse DQU-query to MCoT-gallery directions are weaker, ranging from 61.77 to 61.86 R@10. Cross compatibility is therefore structured and directional, not an automatic consequence of pairing arbitrary endpoint outputs.

Compatibility intervention. Table 4 reports the signed-permutation test from Sec. 3. The maximum change in any preserved endpoint score is 7.75×10^{-7} on FashionGen and 3.58×10^{-7} on FashionIQ. Thus each endpoint and

Table 3. **CIRR validation results in one aligned pipeline.** Subset denotes the six-image candidate group. The best result in each column is bold.

Method	R@1	R@5	R@10	R@50	Subset R@1	Subset R@2	Subset R@3	Avg
MCoT, same pipeline	54.94	85.27	91.89	98.49	78.02	91.94	96.68	81.64
CrossPath, $\alpha = 0.25$	55.23	85.36	92.32	98.40	79.02	92.08	96.99	82.19
CrossPath + target refinement	55.78	85.77	92.42	98.40	79.00	92.15	96.99	82.38

Table 4. **Coordinate-scrambling control.** A shared signed permutation preserves all within-endpoint dot products and diagonal rankings, while breaking cross-endpoint coordinates.

Dataset and path	R@1	R@10	R@50
FashionIQ cross	–	64.58	82.82
FashionIQ scrambled	–	0.85	3.08
FashionGen cross	44.10	88.16	–
FashionGen scrambled	0.14	0.95	–

Table 5. **Additional CrossPath cost.** Gallery storage is for two FP32 embedding sets. Query time excludes offline gallery encoding and gallery dot products.

Setting	Params	Gallery	Time/query
FashionIQ cross mean	0	22.45 MiB	55.44 ms
FashionGen joint	0.526M	41.34 MiB	4.48 ms [†]

[†]Post-embedding router and policy time on an RTX 4090.

the diagonal ensemble retain their rankings up to floating-point precision. Cross-path performance nevertheless collapses from 64.58/82.82 to 0.85/3.08 on FashionIQ and from 44.10/79.74/88.16 to 0.14/0.59/0.95 on FashionGen. The useful signal is carried by aligned cross-endpoint coordinates.

Rescue and regression. At the principal cutoffs, cross paths rescue more base failures than they harm base successes. FashionGen has 294 rescues versus 170 regressions at rank 1, 169 versus 126 at rank 5, and 125 versus 88 at rank 10. FashionIQ has 218 versus 170 at rank 1, 199 versus 138 at rank 10, and 121 versus 92 at rank 50. This balance explains why a simple mean is effective and why cutoff-aware routing provides an additional gain on FashionGen.

4.4. Efficiency

The FashionIQ cross mean adds no trainable fusion parameter. Its 55.44 ms query time is the sum of MCoT and DQU query encoding; gallery representations are computed once. The FashionGen router adds 0.526M parameters and 4.48 ms per query after embedding. It examines only 0.37, 2.02, and 3.85 boundary candidates per query on average at ranks 1, 5, and 10. The method returns a single ranking and does

not require repeated retrieval passes.

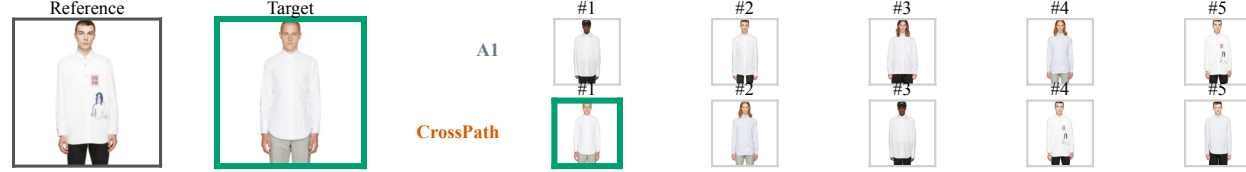
4.5. Qualitative Results

Figure 4 illustrates the quantitative trend on all three benchmarks. The base endpoint often retrieves visually plausible images that retain dominant reference content but miss the requested edit. Cross paths promote targets that better satisfy both the retained appearance and the modification. Cases are selected deterministically by the largest base-miss to CrossPath-hit rank improvement, with no manual visual screening. The figure pipeline reads the original benchmark files; no image is generated or upsampled.

CrossPath retrieval examples across three benchmarks

FashionGen · Case 1 · target rank 9 → 1

Text: "Front: remove graphics, change to hidden placket, and add a small lower-hem logo patch. Back: introduce a center box pleat below the yoke."



FashionIQ · Case 2 · Dress · target rank 19 → 3

Text: "has less sleeves with red and gray / is blue with a higher neckline and black belt"



CIRR · Case 3 · target rank 81 → 4

Text: "Bigger dog, closer to the camera"



Figure 4. **Retrieval examples across FashionGen, FashionIQ, and CIRR.** Each query shows the reference, modification, target, and top-five results from the base endpoint and CrossPath. Green borders mark the target.

5. Conclusion

We presented CrossPath, a complete compatibility view of endpoint composition for composed image retrieval. Two trained endpoints define four retrieval paths, not two. Evaluating the off-diagonal paths reveals combinations of query understanding and gallery geometry that matched endpoint scores cannot express. A parameter-free cross mean improves strong heterogeneous systems on FashionIQ, while lightweight joint routing converts complementary early and deeper rank behavior into consistent FashionGen gains. Conservative fusion and target refinement extend the same principle to open-domain CIRR. The compute-matched diagonal comparison, directional matrix, and coordinate-scrambling intervention identify cross-endpoint compatibility as a distinct source of retrieval ability. This perspective is model agnostic and can accompany future advances in CIR endpoint design without changing their training objectives.

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